Gopi Krishna

Senior AI Engineer

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Professional Summary

- Senior AI Engineer with 5 years of experience designing and deploying cutting-edge LLM-driven systems, specializing in semantic search, Vector embedding, Vector databases, and cloud-native AI architectures.
- Built an AI-powered grant proposal generator using Generative AI (GPT-4), LangChain, and LLM orchestration, integrating chunked and embedded grant data (BAAI, OpenAI) stored in Qdrant/ChromaDB for high-precision semantic retrieval.
- Engineered LangChain workflows for section-wise proposal generation and user-feedback-driven refinement loops, enabling automated structured document creation.
- Designed scalable backend services with FastAPI and WebSocket, deployed on AWS EC2, Lambda, S3, and DynamoDB for secure, high-availability AI operations.
- Developed vector search and semantic query pipelines leveraging Sentence Transformers, GPT-4, LLaMA 3, and Vertex AI Search to enhance LLM-driven retrieval relevance.
- Automated data ingestion and preprocessing using APIs, BeautifulSoup, and Playwright to generate high-quality embeddings for Generative AI workflows.
- Developed scalable, AI-enabled enterprise solutions by creating Python-powered REST APIs, integrating Vertex AI Search, autosuggestions, and intent-driven SQL generation to workflows to streamline data processing and query resolution.
- Integrated Vertex AI for intelligent query processing, function calling, and dynamic SQL generation, and by moving legacy systems to Python/Flask, we engineered a contemporary AI-powered search and data retrieval platform that significantly improved data accuracy and reduced response times.
- Delivered impactful dashboards and visualizations to communicate complex insights and enable data-driven decision-making across teams.

Technical Skills

- **Generative AI & Agentic AI:** LangChain, A2A, GPT-4, LLaMA 3, Sentence Transformers, Hugging Face Transformers, diffusers, vLLM, DSPy, LangGraph, DeepSpeed
- **Deep Learning & Model Architectures:** Vision-Language Models (VLMs), GANs, VAEs, Diffusion Models, Transformers, LoRA Adapters, Model Distillation, Fine-Tuning, RLHF, Instruction Tuning
- **NLP & CV Toolkits:** NLTK, spaCy, torchvision, Ray
- **Programming & Libraries**: Python (Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, PySpark, Jupyter), R, SAS, Java
- **Databases:** SQL, MongoDB, Oracle, MySQL, Snowflake
- **Machine Learning & Algorithms:** Predictive Modeling, Supervised/Unsupervised Learning, Anomaly Detection, Feature Engineering, LLMs, KNN, Regression (Linear, Logistic, Multiple), Naive Bayes, Random Forest, SVM, NLP, K-Means
- **Dataset Curation & Preprocessing:** Large-scale Multimodal Dataset Processing, Synthetic Data Generation, Annotation Workflows, Data Validation
- **Cloud & Deployment:** AWS (Glue, Redshift, SageMaker, Lambda, Athena, S3, EMR, EC2, EKS, ECS, Kinesis, Firehose, IAM), Azure (Web Apps, DevOps), GCP (Vertex AI, Datastore), Databricks
- **GPU Optimization & Distributed Systems:** Mixed-precision training, distributed training strategies, Kubernetes, Docker
- **Testing & Experimentation:** A/B Testing, Cross Validation Strategies, Statistical Hypothesis Testing, Performance Testing, Integration and End-to-End Testing, ML Model Behavior Analysis
- **Big Data & Workflow Automation**: Apache Airflow, Spark, Hadoop, Kafka, Hive, ETL/ELT Pipelines
- **CI/CD & MLOps:** Jenkins, GitHub Actions, Model Deployment Pipelines, MLflow.
- **Data Visualization:** Tableau, Looker Studio, Power BI, Power Pivot
- **Project Management:** Workflow Orchestration, Agile, SDLC, WBS, Slack, Jira
- **Version Control:** Git, GitHub, GitLab, Bitbucket

Experience

University of New Haven **Jan 2025 - present**

AI Research Intern, SAIL LAB Grant Writing Assistant - AI-Powered Grant Finder & Writing Tool

- Scraped data from top funding sources like NIH, NSF, DoD, DoE, and various web platforms, and stored it in Qdrant VectorDB. This allowed us to perform 95% more accurate semantic searches, ensuring better grant matching for researchers.
- Built and integrated a Retrieval-Augmented Generation (RAG) framework using LangChain and LangGraph, which allowed us to combine real-time document retrieval with AI-generated content to help researchers, reducing the manual grant writing time by 60% and improving the proposal relevance scores by 25% based on researcher feedback.
- Developed an autonomous Agentic Bot using GPT-4 and LangChain, making the grant proposal writing process more intuitive. The bot automatically adapts to researchers' inputs, offering personalized suggestions based on the data stored in Qdrant VectorDB, achieving a 90% User Satisfaction Score in pilot testing with 30+ researchers.
- Utilized Qdrant and ChromaDB for semantic search and vector-based retrieval, organizing the scraped funding data for easy and

accurate grant matching, ensuring researchers receive the most relevant opportunities, improving retrieval precision by 20%.

- Amplified the RAG pipeline using Sentence Transformers and BAAI/bge-base-en-v1.5 embeddings. Achieved 18% increase in semantic alignment with the funding agency guidelines and accurately reflected the data in Qdrant Vector DB.
- Designed and deployed a scalable backend system using FastAPI, WebSocket, and DynamoDB to ensure fast, secure access to funding matches. This architecture supports real-time proposal drafting, allowing researchers to interact seamlessly with the system and handle the 300+ concurrent users with 99.9% uptime.
- Built a user-friendly frontend with React.js and Next.js, where researchers can easily input project details, browse matched funding opportunities, and refine AI-generated proposals, all backed by data stored in VectorDB.
- Led the evaluation of search-enabled LLM APIs vs. curated grant databases, comparing their effectiveness in querying the scraped data stored in VectorDB to provide the most accurate and relevant funding opportunities for researchers.

Infosys Pvt Ltd **Oct 2021 - Jul 2023**
AI/ML Engineer Hyderabad, India

- Developed and maintained 10+ scalable Python-based REST APIs using the Flask framework, facilitating seamless integration across enterprise platforms and underpinning robust data workflows.
- Engineered an enterprise search system with Vertex AI Search using data from Google Cloud Datastores, significantly improving data retrieval accuracy by 42% and reducing response time.
- Designed and implemented intelligent autosuggestion features for typeahead inputs with the Vertex AI Suggestions client and custom logic to enhance query understanding.
- Enhanced natural language query handling by integrating function calling and intent detection, utilizing few-shot learning to generate SQL queries dynamically and revamped data pipeline performance by 25%.
- Improved retrieval quality by applying custom re-ranking expressions on Vertex AI Search results, actively troubleshooting technical issues to boost data consistency and relevance.
- Managed migration of 7+ legacy C# search modules to modern Python-based implementations with Vertex AI clients, ensuring consistent performance and streamlined integration speed of data processes by 40%.
- Collaborated on Azure DevOps workflows by managing pull requests, conducting code reviews, and overseeing deployments to Azure Web Apps to support scalable and efficient project delivery.

Wipro Ltd **Jun 2019 - Oct 2021**
Data Analyst Hyderabad, India

- Processed large datasets to extract actionable insights and generate clear, impactful visualizations using Python and Power BI, making complex data accessible to non-technical stakeholders.
- Developed Power BI dashboards for shortage analysis, simplifying complex metrics and improving visibility for business users.
- Engineered comprehensive Power BI visualizations to convey intricate data trends, improving stakeholder understanding by 25% and increasing project approval rates by 10%.
- Maintained 98% data pipeline stability across various visualization workflows, reducing discrepancies by 20% over a year.
- Collaborated with a cross-functional team of over 5 data engineers and analysts to build and deploy 10+ end-to-end data visualization solutions.
- Conducted Power BI training sessions and shared best practices, leading to a 20% improvement in team-wide efficiency for data visualization and reporting tasks.
- Delivered ad-hoc reports and analytical insights to cross-functional teams, enabling faster, data-driven decision-making and reducing turnaround time by 20%.
- Performed in-depth data audits to identify and correct inconsistencies, enhancing data integrity by 15% and supporting more accurate business reporting.
- Effectively communicated complex analytical findings through accessible reports and presentations tailored for both technical and non-technical audiences.

Education

University of New Haven
MS, Data Science

- **Coursework:** Mathematics for Data Science, Introduction to Artificial Intelligence, Exploratory data analysis, Machine Learning, Power BI Dashboarding, Natural Language Processing, Deep Learning, Leadership and Entrepreneurism

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional
- Oracle AI Vector Search Certified Professional
- AWS Certified AI Practitioner.
- AWS Certified Cloud Practitioner.
- Infosys Certified Agile Developer.
- AWS Machine Learning Specialty in progress.